

Phase I/II Operating Characteristics

Continuous enrollment | 56-day arrival | IPDE alpha = 0.3

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	9.4	4.6	2.3	0.9	0.3	82.5
C2 alpha=0.3 one-stage	10	3.5	1.8	0.5	0.1	84.1
C3 alpha=0.3 one-stage	9.7	3.7	1	0.3	0.1	85.1
Oracle C1 alpha=0 two-stage	9.2	5.1	2.2	0.8	0.3	82.4
C2 alpha=0.3 two-stage	9.8	4.2	1.7	0.6	0.1	83.6
C3 alpha=0.3 two-stage	9.4	4.3	1.1	0.2	0.1	84.9
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	11	4.2	2.3	0.9	0.2	
C2 alpha=0.3 one-stage	11.4	3.9	1.8	0.6	0.2	
C3 alpha=0.3 one-stage	11.6	3.8	1.7	0.4	0.1	
Oracle C1 alpha=0 two-stage	10.9	4.3	2.4	0.9	0.3	
C2 alpha=0.3 two-stage	11.3	4	1.9	0.6	0.2	
C3 alpha=0.3 two-stage	11.5	3.9	1.7	0.4	0.1	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	23.9	21	19.3	21.8	13.1	0.9
C2 alpha=0.3 one-stage	25.5	23.7	18.9	19.6	12.1	0.3
C3 alpha=0.3 one-stage	23	25.5	19.3	19.8	12.1	0.3
Oracle C1 alpha=0 two-stage	24.1	21.9	18.3	21.1	13.7	0.9
C2 alpha=0.3 two-stage	25.4	26.6	18.1	18.9	10.8	0.3
C3 alpha=0.3 two-stage	24.6	26.6	19.4	18.7	10.5	0.3
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	7.2	6.7	5.8	5.8	4.5	
C2 alpha=0.3 one-stage	7.3	7	5.6	5.7	4.4	
C3 alpha=0.3 one-stage	7.2	7.2	5.8	5.5	4.3	
Oracle C1 alpha=0 two-stage	7.1	6.7	5.7	5.8	4.8	
C2 alpha=0.3 two-stage	7	7.1	5.7	5.8	4.5	
C3 alpha=0.3 two-stage	7	7.2	5.8	5.6	4.5	

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Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	4.9	47	28.4	12.3	2.2	5.2
C2 alpha=0.3 one-stage	5.1	50.7	28.1	9.1	1.3	5.7
C3 alpha=0.3 one-stage	5.4	49.2	29.4	9.3	1	5.6
Oracle C1 alpha=0 two-stage	4.5	48.5	29.1	10.9	1.9	5.1
C2 alpha=0.3 two-stage	5.6	51.4	27.2	9.2	1.5	5.1
C3 alpha=0.3 two-stage	5.4	52	28.6	7.4	1.6	5
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	5.2	11.3	7.6	4.1	1.3	
C2 alpha=0.3 one-stage	5.5	11.6	7.4	3.9	1	
C3 alpha=0.3 one-stage	5.4	11.7	7.7	3.7	0.9	
Oracle C1 alpha=0 two-stage	5.1	10.7	8	4.3	1.4	
C2 alpha=0.3 two-stage	5.3	10.9	7.8	4.1	1.3	
C3 alpha=0.3 two-stage	5.2	11	7.9	4	1.2	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	0.8	1.4	6	59.4	32.4	0
C2 alpha=0.3 one-stage	0.6	1.1	3.2	61.7	33.4	0
C3 alpha=0.3 one-stage	0.6	1.2	3.1	62.9	32.2	0
Oracle C1 alpha=0 two-stage	1.1	1.2	6.3	56	35.4	0
C2 alpha=0.3 two-stage	1	1	4.4	55.2	38.4	0
C3 alpha=0.3 two-stage	0.9	1.2	4.9	54.4	38.6	0
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	3.4	3.5	4.5	12.5	6.2	
C2 alpha=0.3 one-stage	3.4	3.4	4.3	12.8	6.1	
C3 alpha=0.3 one-stage	3.4	3.4	4.3	12.9	6	
Oracle C1 alpha=0 two-stage	3.4	3.4	4.3	10.9	8	
C2 alpha=0.3 two-stage	3.4	3.3	4.2	10.9	8.2	
C3 alpha=0.3 two-stage	3.4	3.4	4.2	10.6	8.4	

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Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	10.7	25.3	44.6	9.5	0.6	9.3
C2 alpha=0.3 one-stage	11.5	23.8	46.4	8.3	0.7	9.2
C3 alpha=0.3 one-stage	11.7	26.1	45.3	7.4	0.5	8.9
Oracle C1 alpha=0 two-stage	11.3	23.8	46.3	9	0.9	8.7
C2 alpha=0.3 two-stage	11.1	26.1	46.1	7.7	0.7	8.2
C3 alpha=0.3 two-stage	11.6	26.2	46.3	6.5	0.7	8.6
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	7.3	8	9.8	3.4	0.6	
C2 alpha=0.3 one-stage	7.5	8.3	9.8	3	0.4	
C3 alpha=0.3 one-stage	7.5	8.4	9.9	2.9	0.4	
Oracle C1 alpha=0 two-stage	6.9	7.9	9.7	3.8	0.9	
C2 alpha=0.3 two-stage	7.2	8.1	9.5	3.5	0.8	
C3 alpha=0.3 two-stage	7.2	8.1	9.6	3.3	0.8	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	3	6.5	8.7	11.8	65.3	4.7
C2 alpha=0.3 one-stage	4.8	7.2	8.4	16	58.8	4.7
C3 alpha=0.3 one-stage	5.3	6.4	10.3	15	58.1	4.8
Oracle C1 alpha=0 two-stage	4.6	7.2	8.6	11.6	63.2	4.8
C2 alpha=0.3 two-stage	5	6.7	9.1	15.8	58.7	4.6
C3 alpha=0.3 two-stage	5	8.8	10.1	14.9	56.5	4.6
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	4.6	4.6	5	5.5	10.2	
C2 alpha=0.3 one-stage	4.7	4.7	5.2	5.7	9.6	
C3 alpha=0.3 one-stage	4.7	4.7	5.4	5.5	9.5	
Oracle C1 alpha=0 two-stage	4.5	4.5	4.9	5.6	10.3	
C2 alpha=0.3 two-stage	4.6	4.7	5.1	5.7	9.8	
C3 alpha=0.3 two-stage	4.6	4.8	5.2	5.5	9.6	

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Scenario	1	16	20	24	27	38	Mean
Oracle C1 alpha=0 one-stage	18.7	30	29.5	30	29.1	29.8	27.8
C2 alpha=0.3 one-stage	11.6	18.2	18.7	19.6	19.1	18.7	17.6
C3 alpha=0.3 one-stage	9.9	14.9	15.9	16.9	16.5	15.6	15
Oracle C1 alpha=0 two-stage	18.8	30	29.5	30	29.1	29.8	27.9
C2 alpha=0.3 two-stage	11.6	18.2	18.8	19.7	19.3	18.8	17.7
C3 alpha=0.3 two-stage	9.9	14.9	16.1	17.1	16.8	15.6	15.1

Values are reported from Total_Unique_Patients; comparator counts without patient IDs are sums of dose-level means.

Rows are one-stage first, then two-stage. C1 is fixed at alpha=0; C2/C3 use alpha=0.3.